

FarmShare

Direct-to-Community Surplus Food Distribution

Team Name: Core4

Problem Statement

Build a platform connecting farmers with surplus produce directly to food banks and NGOs to reduce food waste.

Objective

Develop a sustainable platform that helps farmers donate or distribute surplus produce to food banks and NGOs, reducing food waste while ensuring excess food reaches people in need.



The Food Waste Gap

Agricultural Excess vs. Nutrition Insecurity

- The Surplus: Tons of fresh, edible farm produce go to waste due to overproduction or cosmetic flaws.
- The Friction: Farmers lack a direct, rapid channel to find organizations that can collect food before it spoils.
- The Need: NGOs and food banks lack real-time visibility into local farm surpluses.
- The Flow: Connects farmers directly to nearby NGOs to claim surplus in 6 clear steps: list, parse, alert, claim, accept, and deliver.

FARMSHARE SURPLUS FOOD DELIVERY PROCESS



HarvestLink Voice AI

Voice-Activated Form Autofill

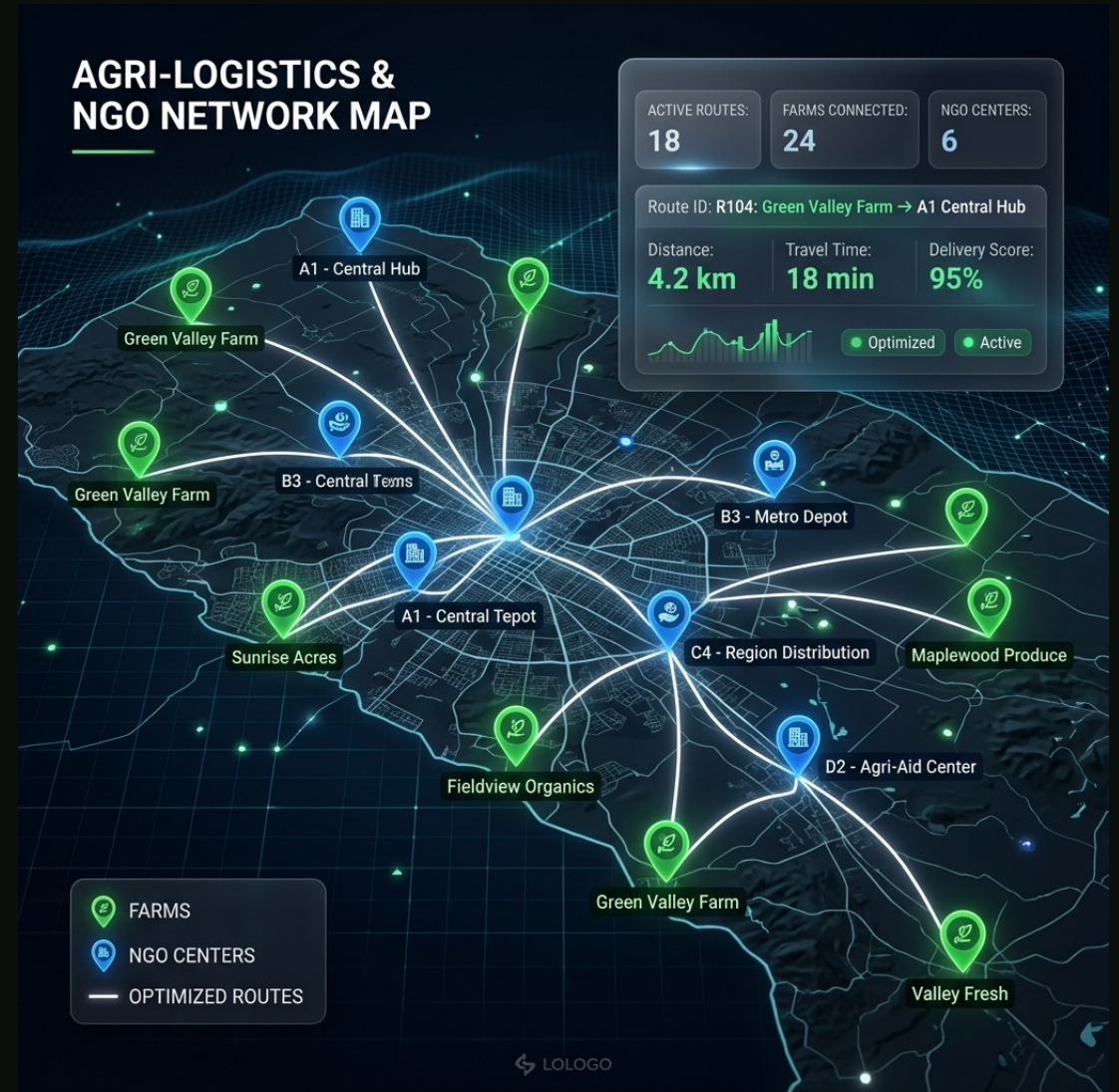
- Multilingual Support: Speaks English, Hindi (■■■■■), Marathi (■■■■■), Telugu (■■■■■), and Kannada (■■■■■).
- Voice-to-Text: Captures farmer speech in the field using standard browser Web Speech API.
- AWS Bedrock Nova Pro: Processes audio transcriptions to parse crop name, quantity, and harvest dates.
- Context Resolution: Translates regional crops to English and resolves words like 'today' to standard dates.



Smart Proximity Logistics

Optimized Routing & Instant Alerts

- **Intelligent Matching:** Our engine calculates logistics scores based on straight-line distance and crop freshness.
- **100km Notification Radius:** Local NGOs receive immediate email alerts with a one-click claim button the moment food is listed.



Closed-Loop Delivery

Full Transparency from Field to Family

- Strict State Control: A clear 4-step pipeline ensures no double claiming. Accepting one NGO auto-rejects competitors.
- Verified Handoffs: Operations are finalized only when transport drivers mark the food as physically collected on-site.



Measuring Impact

Dashboard Statistics & Real-Time Analytics

- Live Aggregation: Every successful delivery updates global metrics, tracking exact kilograms of food rescued.
- Join The Network: FarmShare is fully scalable and ready to empower your community's agricultural logistics.

FOOD RESCUE IMPACT DASHBOARD

PREMIUM PRESENTATION ASSET | Q4 2023



Overview

Metrics

Partners

Reports

Admin | John Doe

5,240 kg

Total Food Rescued
(Kilograms this quarter)



48

Verified Farmers
(Active agricultural partners)



18

Local NGOs
(Distribution non-profits)



FOOD WASTE REDUCTION TRENDS

Cumulative reduction in waste landfilled (Kg)

Kg

5000

4000

3000

2000

1000

0



KEY ACHIEVEMENT

- 🌱 Cumulative reduction in food rescue operations
- 🌱 Continuous reduction in residual food waste
- 👤 Proven reduction in waste landfilled

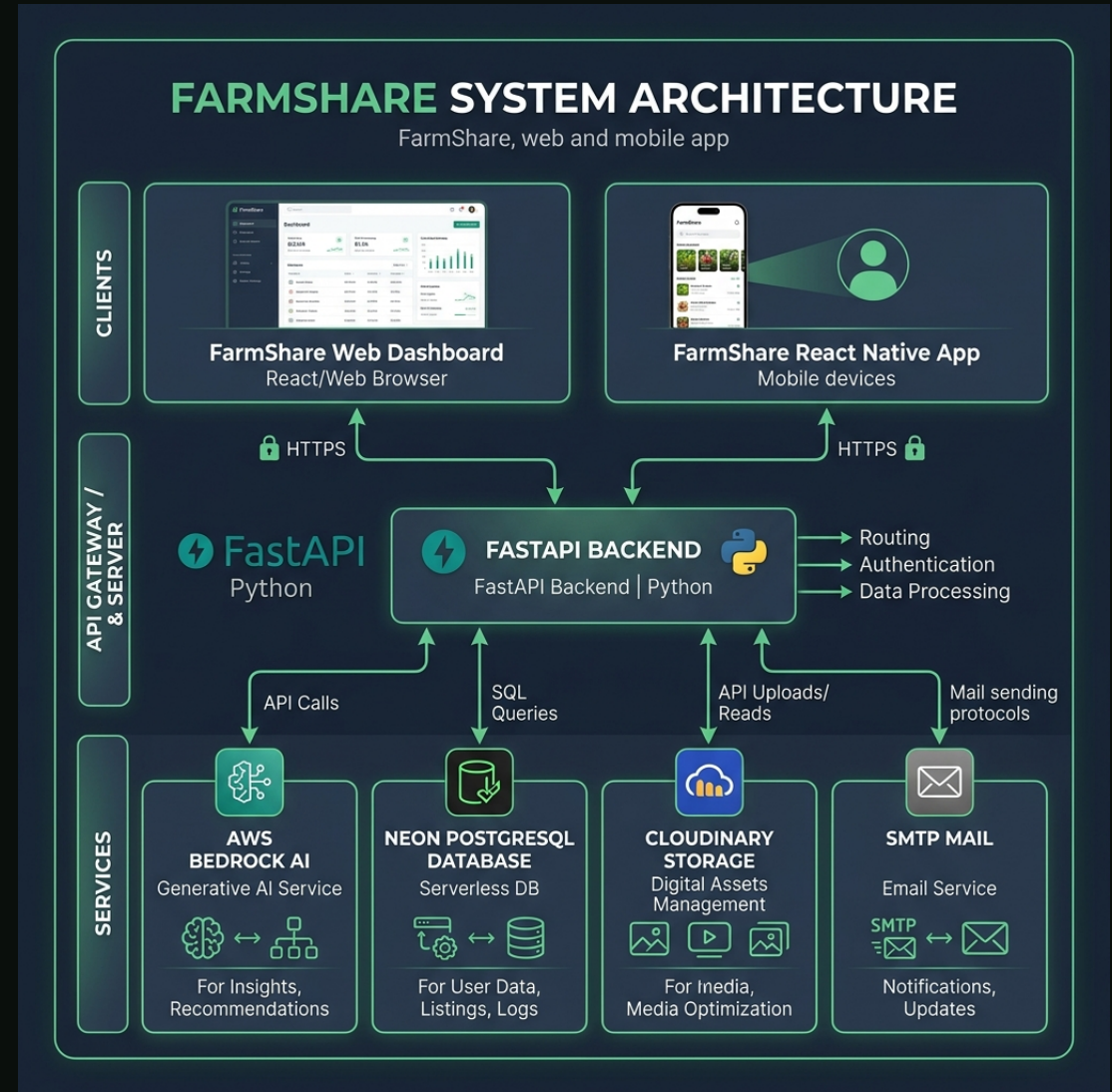
SF Pro Display

March 2024

Backend Architecture Map

Secure & Scalable FastAPI Ecosystem (No Redis)

- Layer 1 (API Gateway): Nginx reverse proxy routes traffic to FastAPI application routers (Auth, Produce, Requests, Stats).
- Layer 2 (Logic Layer): Manages JWT validation, Smart Match distance score logic, and background email alert tasks.
- Layer 3 (Services): Neon serverless database, AWS Bedrock AI inference, Cloudinary media storage, and SMTP alerts.



Tech Stack & Frameworks

Lightweight, High-Performance Infrastructure

- Backend System: FastAPI, Python 3.10+, Neon Serverless PostgreSQL, raw psycopg2 query pool.
- Frontend Stack: HTML5, CSS Glassmorphism, Leaflet.js, OpenStreetMap.
- Mobile App: React Native (Expo framework), React Navigation, WebViews for identical maps rendering.
- API Services: AWS Bedrock, Cloudinary Image Storage, SMTP Mail dispatcher.

